MethylPhys CPG - Operations Manual, chain v4

The first commissioned IAM chain. A research instrument, not a diagnostic test. Generated 2026-10-18 by manual/build_manual_v4.py from the files named at the start of each part.

Part 1. The procedure - chain v4

Source: `sop/MethylPhys_CPG_SOP_v4.md`

MethylPhys CPG SOP: chain v4

Generated by `build_sop_v4.py` from the chain's own files. Do not edit by hand. Where this document and the code differ, the code is what runs.

The chain's job: one sample goes in; every commissioned instrument reads it inside its commissioned scope, reads outside it with a plain note saying it has not yet been tested there, or declines with its reason. Research use only; not a diagnostic test.

1. What is commissioned

Instrument and scope	Commissioned	Note
Met-A, isolated neutrophils, EPIC v1	2026-10-09	COMMISSIONING_NOTE_METAA_NEUTROPHILS.md
Met-A C-score, whole blood, EPIC v1	2026-10-15	COMMISSIONING_NOTE_CSCORE.md
IAM-Atlas v2.1, per use	2026-10-16	COMMISSIONING_NOTE_ATLAS_V21.md
Deconvolution tool, blood-cell shares, arrays	2026-10-17	COMMISSIONING_NOTE_DECONV_V21.md
IAM-A, healthy blood CD8 T cells, sequencing	2026-10-17	COMMISSIONING_NOTE_IAMA.md

Each was commissioned by a test sealed before its data were read. A reading outside these scopes is printed with the note "This score (or share) is preliminary until ... is officially commissioned on ..." and is not a commissioned result.

2. Install

From a fresh clone of the repository, Python 3.11:

```
git clone https://github.com/hmahaffeyges/IAM-Validation.git && cd IAM-Validation/Biological_Physics/MethylPhys/cha
python3 -m venv .venv && . .venv/bin/activate && pip install -r requirements.txt
```

Stage 1 needs the Illumina manifest; methylprep downloads it on first use into `$HOME/.methylprep_manifest_files/`. Without it the run stops with `ENVIRONMENT_MISSING_MANIFEST` (exit 3) and the array is not judged.

3. Check the chain (one command)

```
python3 release_check_v4.py
```

It runs every check of the chain, including real arrays from the chain's test data, planted samples for each instrument, and the checksum of every frozen file below. Every check must pass; a run that does not pass is not a commissioned chain.

4. Read one sample

Arrays (a raw IDAT pair, Illumina EPIC v1):

```
cd MethylPhys_Interface
python3 run_sample.py --grn S_Grn.idat --red S_Red.idat --specimen "whole blood" --id S001 --out S001.html
```

Isolated neutrophils: `--specimen "isolated neutrophils"`. Optional: `--sex F|M`, `--age`, recorded when given. Same-run healthy references for the median tare: `--ref-table refs.csv`, three or more healthy arrays

read on the same run, with column `A` (their Met-A reading) and column `C` (their C-score), each from that array's own report bundle.

Sequencing (whole-genome bisulfite), against the sample's own reference:

```
python3 run_sample.py --pat S.pat.gz --seq-cell "cd8 t cells" --id S002 --out S002.html
python3 run_sample.py --bam S.bam --ref-fasta hg19.fa --assembly hg19 --seq-cell "cd8 t cells" --id S003 --out S003
```

`--seq-cell` names the cell type; the sample-shape check has a healthy shape for 56 cell types: adipocytes, aorta endothelium, b cells, bladder epithelium, breast basal epithelium, breast luminal epithelium, cardiomyocyte, cd4 t cells, cd8 t cells, colon enteroendocrine, colon epithelium, colon fibroblasts, colon macrophages, cortical neurons, effector memory cd8 t cells, endometrium epithelium, erythrocyte progenitors, fallopien tubes epithelium, gastric antrum epithelium, gastric body epithelium, gastric fundus epithelium, heart fibroblasts, hepatocyte, kidney glomerular endothelium, kidney glomerular epithelium, kidney tubular endothelium, kidney tubular epithelium, lung alveolar endothelium, lung alveolar epithelium, lung alveolar macrophages, lung bronchus epithelium, lung interstitial macrophages, memory b cells, monocytes, naive cd4 t cells, naive cd8 t cells, nk cells, oligodendrocytes, pancreas endothelium, pancreatic acinar, pancreatic alpha, pancreatic beta, pancreatic delta, pancreatic duct, prostate epithelium, small intestine epithelium, smooth muscle, striated muscle, t central memory cd4, t effector cell cd8, t effector memory cd4, thyroid epithelium, tongue epithelium, tonsil palatine epithelium, tonsil pharyngeal epithelium, vascular saphenous endothelium.

5. What the chain declines, and why

- An EPIC v2 or 450K array: refused at intake with one sentence (no frozen Met-A reference for that platform).
- A failed array (Stage 0 quarantine): exit 2, no report.
- A sequencing sample whose shape is outside the healthy shape of its declared cell type, or with too few qualifying molecules: IAM-A declines and says why.
- A noisy array (the noise gate): the Met-A state is withheld and no gauge is drawn.

6. Outputs

- `S001.html`: the report. At a glance; cell shares (IAM-Deconvolver against IAM-Atlas v2.1); Met-A; the C-score with its HEALPix sky plate; IAM-A; the technical record.
- `S001_bundle.json`: every stage's output, with the chain's commit.
- one row appended to `evidence_ledger.jsonl` beside the report.

7. Stages in the live path

32 steps, in order (from `chain/chain_sequence.json`):

Step	Code
specimen_refusal	`stage_0_intake.py`
step_0_1_idat_arrival	`stage_0_intake.py`
step_0_2_manifest_creation	`stage_0_intake.py`
step_0_3_integrity_hash	`stage_0_intake.py`
step_0_4_control_probe_validation	`stage_0_intake.py`
step_0_5_detection_pvalue_qc	`stage_0_intake.py`
step_0_6_bead_count_qc	`stage_0_intake.py`
step_0_7_call_rate	`stage_0_intake.py`
step_0_7b_platform_coverage	`stage_0_intake.py`
step_0_8_sex_check	`stage_0_intake.py`
step_0_9_decision_gate	`stage_0_intake.py`
Stage 1 - IDAT calibration	`stage_1_idat_calibration.py`

Step	Code
platform_refusal	conductor_v3.py
stage_t_selftare_ii	conductor_v3.py
stage_m_isolated	conductor_v3.py
stage_a_composition	conductor_v3.py
stage_m_blood	conductor_v3.py
noise_index	conductor_v3.py
stage_t_tare	conductor_v3.py
noise_gate	conductor_v3.py
stage_mc_cscore	conductor_v3.py
stage_t_cscore	conductor_v3.py
stage_v4.stage_d21	stage_v4.py
stage_v4.cscore_sky	stage_v4.py
stage_q0_intake.intake	stage_q0_intake.py
stage_v4.stage_q2	stage_v4.py
stage_q_iam_a.pat_site_table	stage_q_iam_a.py
stage_q_iam_a.read	stage_q_iam_a.py
stage_q_iam_a.cscore	stage_q_iam_a.py
serial_mode.check_same_person	serial_mode.py
serial_mode.delta_sky	serial_mode.py
Report	MethylPhys_Interface/report_v4.py

8. Frozen inputs

Every reference the chain reads is a committed file; nothing is fitted at run time. The release check verifies each checksum.

File (in chain/)	SHA-256
../atlas/v2.1/compact_manifest.json	fafb600b4545dabb…
Runtime Matrices/Gauge/cell_gauge_marks_v1.json	d23916d54a8fe7b0…
Runtime Matrices/IAM_A_SelfRef/band_v1.json	d4c32f760d98ad14…
Runtime Matrices/IAM_A_SelfRef/selfref_line_v1.json	540157559b9e27bb…
Runtime Matrices/IAM_A_SelfRef/shape_reference_v1.json	41470faf73551699…
Runtime Matrices/Intake/intake_thresholds_v1.json	58f7e6073e8f2a5c…
Runtime Matrices/Met_A_Floors/blood_composition_EPIC_v1.json	6ad3dfb0e01a70ee…
Runtime Matrices/Met_A_Floors/metA_floors_v1_3.json	2f64f8846f8a4b0d…
Runtime Matrices/Met_A_Floors/metA_floors_v1_3_loo.csv	73f090a7004bc7c7…
Runtime Matrices/Met_A_Floors/neutrophil_reference_v1_2.json	820c9b1ecf824561…
Runtime Matrices/Met_A_Floors/noise_gate_EPIC_v1.json	825f64f7b6c7077a…
Runtime Matrices/Met_A_Floors/noise_sites_EPIC_v1.json	3531561783261c57…

9. Rules

- Synthetic first: every new test runs on planted data, read by the chain's exact rule, before any real data is opened.
- Sealed before read: a test's samples, rule and bars are committed before its data is downloaded; it is scored once.

- Clean clone: every chain change passes `release_check_v4.py` on a fresh clone, on a clean machine, before it is called tested.
- One reading per instrument; no lines beyond Normal are printed until they are measured on the gauge.

The earlier procedure, for chain v3, is kept as the record of how the chain was built: MethylPhys_CPG_SOP_v3.md.

Part 2. Operations - chain v4

Source: `manual/OM_v4_chain.md`

Operations Manual: chain v4 (operator chapter)

The procedure (install, the one-command check, the commands, what is declined, outputs, frozen inputs) is the SOP, printed first in this manual. This chapter is what an operator needs around it: before a run, reading the report, and what to do when the chain stops.

Before a run

1. Run the release check once per installation and after every update: `python3 release_check_v4.py`. Every check must pass.
2. Arrays: the Illumina manifest must be in `$HOME/.methylprep_manifest_files/` (methylprep fetches it on first use). Read the first array of a batch alone, so the download is not raced.
3. Same-run references (arrays): read three or more healthy arrays from the same run first, each on its own. Put their Met-A reading (`A`) and C-score (`C`) from each report's bundle in a CSV with columns `id,A,C`, and pass it with `--ref-table` when reading the samples of that run. The tare divides by the references' median, so the laboratory's own habits cancel out. Nothing is fitted.
4. Sequencing: give the cell type with `--seq-cell`, and for a BAM the exact reference FASTA it was aligned to (`--ref-fasta`, indexed).
5. Declare the specimen exactly (`--specimen "whole blood"`, `"isolated neutrophils"`, `"buffy coat"` or `"constructed DNA mixture"`). Anything else is refused at intake with its reason.

Reading the report

Each run writes one HTML page (a Researcher view, opened first, and a Plain view), the same report as two PDFs, the bundle (JSON) and one row in the evidence ledger. The report reads top to bottom:

- **At a glance.** One line per instrument: its reading, its state, and whether the reading is inside that instrument's commissioned scope.
- **Every gauge has three marks:** 0, the physical floor; 1, healthy; and the saturation end, where every identity site is fully scrambled (Met-A 3.03; IAM-A 0.5 / ε_pred(L), the sample's own). The marks are derived, not fitted.
- **States.** Met-A, tared: Normal 0.95 to 1.05. The C-score: above its healthy band when above 1.152 (one-sided). IAM-A: Normal 0.7787 to 1.0551. Where these bands mean disease is for cancer researchers to find; the report prints readings and states, not diagnoses.
- **Commissioned scope.** A reading outside what an instrument was commissioned on is still shown, with the line *This score is preliminary until [instrument] is officially commissioned on [specimen or cell]*. Met-A is commissioned on isolated neutrophils, the C-score on whole blood, the IAM-Deconvolver on neutrophil, monocyte, CD4 T and CD8 T shares, IAM-A on healthy CD8 T cells.
- **Cell shares.** Each cell's share with its 95 % interval, read against IAM-Atlas v2.1. Before mixing anything, the tool checks the sample can be read; it refuses rather than report a wrong share.
- **The tare.** When `--ref-table` is given, the report prints the references' median and spread, the tared reading, and the smallest change this sample could have shown.
- **The noise gate.** On a noisy array Met-A's state is withheld and no gauge is drawn; the report says so.
- **The sky.** The sample laid over healthy neutrophils from IAM-Atlas v2.1 on one HEALPix sphere, every site in its genome place. Grey is where the sample matches healthy; color marks only departures beyond twice the sample's own spread. Scattered color is what healthy looks like; streaks along the genome are what the C-score calls clustered.

- **IAM-A** reads a sequencing sample against itself: the error with which its own cells hold their marks, against the healthy line of its declared cell type. If the sample's shape is outside that cell type's healthy shape, or too few molecules qualify, IAM-A declines and says why.
- **Technical record.** The chain's commit, every input's checksum, every frozen file read, and every caution (for example, Stage 0 did not run because the sample entered as a table of values).

When the chain stops

exit	printed	meaning	what to do
0	the report	read, or declined with a reason printed in the report (specimen, platform, sample shape)	read the report
1	an argument message	a command-line input is missing or malformed (for example a reference table without its columns)	correct the command
2	`QUARANTINE` and its code	Stage 0 failed the file (corrupt or truncated IDAT, missing channel, array type mismatch, low detection)	nothing is scored; check the file, rerun from the scanner output
3	`ENVIRONMENT_MISSING_MANIFEST`	the Illumina manifest could not be fetched	put the manifest in place and rerun; the array was not judged

An EPIC v2 or 450K array is refused with one sentence (`PLATFORM_REFUSED`); the chain holds no Met-A reference for those platforms.

Reproducing the commissioning

The commissioning tests and the headline results rebuild from public data with one command each: `reproduce/README.md` at the top of the repository.

Part 3. The chain, step by step

Source: doors/CHAIN_SEQUENCE.md

The chain, step by step - derived from the code

Generated by chain/build_chain_sequence.py from the AST of run_sample.py, conductor_v3.py and stage_q_iam_a.py. Do not edit by hand: re-run the generator. Every step below is a call the code actually makes, in the order it makes it. Chain v3 is the only engine; the class-floor engine (v2) was retired on 2026-10-03 and is archived privately.

The live path (chain v4) - 32 steps, in order

#	step	implemented in	what it does
1	specimen_refusal	stage_0_intake.py	None when intake accepts the specimen (ACCEPTED_SPECIMENS); otherwise the plain refusal text naming the specimen (decision L)
2	step_0_1_idat_arrival	stage_0_intake.py	SOP section 0.1_idat_arrival
3	step_0_2_manifest_creation	stage_0_intake.py	SOP section 0.2_manifest_creation
4	step_0_3_integrity_hash	stage_0_intake.py	SOP section 0.3_integrity_hash
5	step_0_4_control_probe_validation	stage_0_intake.py	SOP section 0.4_control_probe_validation
6	step_0_5_detection_pvalue_qc	stage_0_intake.py	SOP section 0.5_detection_pvalue_qc
7	step_0_6_bead_count_qc	stage_0_intake.py	SOP section 0.6_bead_count_qc
8	step_0_7_call_rate	stage_0_intake.py	SOP section 0.7_call_rate
9	step_0_7b_platform_coverage	stage_0_intake.py	SOP section 0.7b_platform_coverage
10	step_0_8_sex_check	stage_0_intake.py	SOP section 0.8_sex_check
11	step_0_9_decision_gate	stage_0_intake.py	SOP section 0.9_decision_gate
12	Stage 1 - IDAT calibration	stage_1_idat_calibration.py	Stage 1 - IDAT calibration to beta (SOP Stage 1, steps 1.1-1.2 + 1.5) _called by run_sample.py when given --grn/--red; skipped when given --betas_
13	platform_refusal	conductor_v3.py	None when the specimen is EPIC v1; otherwise the refusal text
14	stage_t_selftare_ii	conductor_v3.py	Stage T step 1, self-tare II (adopted 2026-10-04, DEV-SELFTARE-02): per probe design, beta' = Lr + (beta - L)(Ur - Lr)/(U - L) from this
15	stage_m_isolated	conductor_v3.py	Isolated / sorted neutrophils against their healthy reference. The healthy-reference state is kept as state_own_floor; the reading's state is
16	stage_a_composition	conductor_v3.py	EPIC blood composition (blood_composition_EPIC_v1): 8 groups from Salas purified EPIC cells; markers exclude the neutrophil sites;
17	stage_m_blood	conductor_v3.py	Whole blood: Met-A = mean H(beta) / mean H(e) at the neutrophil sites, e = sum_g f_g mu_g (EPIC purified group profiles)

#	step	implemented in	what it does
18	`noise_index`	`conductor_v3.py`	N = mean H(beta) over the noise sites measured on this array; None when fewer than MIN_NOISE_FRACTION are measured _stage 9: the array's noise index N over the noise sites_
19	`stage_t_tare`	`conductor_v3.py`	Same-run tare against >= MIN_REFS healthy references of the same specimen type run the same way (same slide, else same batch)
20	`noise_gate`	`conductor_v3.py`	Runtime Matrices/Met_A_Floors/noise_gate_EPIC_v1.json: N_max = top of the reference arrays' noise range (DEV-NOISE-01) _stage 9: N_max read here; run_neutrophil withholds the gauge state when N > N_max and the reading is untared_
21	`stage_mc_cscore`	`conductor_v3.py`	Met-A C-score (stage 6): clustering of the residual z map in genomic order - variance of the means of blocks of clustering_block
22	`stage_t_cscore`	`conductor_v3.py`	Same-run tare of the Met-A C-score, the same rule as the median tare of A (DEV-CSCORE-TARE-01): C_rel = C / median(C of >= MIN_REFS
23	`stage_v4.stage_d21`	`stage_v4.py`	IAM-Deconvolver: cell shares against IAM-Atlas v2.1 _array input_
24	`stage_v4.cscore_sky`	`stage_v4.py`	z: the C-Score's per-site departure map (pd.Series by site, in the C-Score's genomic order). Writes a Mollweide view; returns a summary _array input; a picture, nothing scored_
25	`stage_q0_intake.intake`	`stage_q0_intake.py`	Stage Q0 on one .pat(.gz). alignment_qc: dict (or JSON path) from the alignment step with 'conversion_rate' and/or _runs when run_sample.py is given --pat, before Stage Q_
26	`stage_v4.stage_q2`	`stage_v4.py`	IAM-A: count, sample-shape check, reading against the sample's own loose share on band v1 _runs when run_sample.py is given --pat or --bam (the retired position reading only with --dev-iama-position)_
27	`stage_q_iam_a.pat_site_table`	`stage_q_iam_a.py`	Per-site table (pos, opp_A, err_A, opp_B, err_B) from one .pat(.gz) file, pipeline loyfer_pat_v1. pos = 'chrom:CpG index' _development: --site-table, or --pat with --dev-iama-position_
28	`stage_q_iam_a.read`	`stage_q_iam_a.py`	site_table: per-site table (see module doc). cell: e.g. 'neutrophils'. pipeline: the read-level pipeline that produced the table _development: --site-table, or --pat with --dev-iama-position_
29	`stage_q_iam_a.cscore`	`stage_q_iam_a.py`	IAM-A C-score for the pooled table (opp = opp_A + opp_B) and for each half. Returns {C, n_blocks, se_null, halves: {...}} _development: --site-table, or --pat with --dev-iama-position_

#	step	implemented in	what it does
30	`serial_mode.check_same_person`	`serial_mode.py`	Refuse unless patient hash, array type and pipeline match. Returns (ok, reason) _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
31	`serial_mode.delta_sky`	`serial_mode.py`	beta_now - beta_prior per address on the addresses both draws measured. No expectation, no sigma, no composition _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
32	`Report`	`MethylPhys_Interface/report_v4.py`	one self-contained HTML page plus the JSON bundle

Stage modules in chain/ that the live path does not call

module	what it implements	status
`stage_2c_trace_detection.py`	Stage 2c - is there any evidence of a trace class in this specimen?	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_4_6_patient_cmb.py`	Stage 4.6 — the patient's sky (PROC-CMB-01, 2026-09-21). Replaces the retired patient_brightness_comparison.py	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_a_composition_v2.py`	Stage A - composition, atlas v2 solver (deconv_v2). Not called by chain v3 (conductor_v3.stage_a_composition uses blood_composition_EPIC_v1.json). Frozen settings (commissioned as PROC-DECONV-V2-01; specimen cell sets 2026-09-30)	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_atlas_commission.py`	Stage A21 - atlas v2.1 commissioning scorers, as AMENDED 2026-10-16 before the atlas exists (author: "Seal the amendment")	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_atlas_u1.py`	Atlas v2.1 use test U1 (reference profiles), scored ONCE by the sealed and amended bar (chain Stage A21), written 2026-10-16 before any	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_atlas_u1_planted.py`	Planted test of score_u1_v21.py (synthetic first): fake blocks with known draws, arrays drawn from the predictive distribution (must pass),	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_atlas_u2.py`	Atlas v2.1 U2a, blind sites of the four sealed tissue tests (sealed_commission_v21.json U2; bar B2 of amendment_v21b.json; chain Stage A21	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_atlas_u3.py`	Atlas v2.1 U3, blood composition (sealed bar B3, amendment_v21b.json; scorer chain Stage A21 u3_neutrophil). Written 2026-10-16 before any	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_deconv_v21.py`	Stage D21 - the deconvolution tool (IAM-Deconvolver) on IAM-Atlas v2.1. Written 2026-10-16; COMMISSIONED 2026-10-17 for the shares of	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_temperature.py`	Chain stage TL: the temperature law on single molecules (PRED-TEMPERATURE-01, PHY-057). Written 2026-10-17, before any read is opened	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)

Behind development flags (DEVELOPMENT - not commissioned; never part of a reading)

flag	step	implemented in	what it does
`--dev-direction`	`dev_stages.direction_record`	`dev_stages.py`	Signed move toward or away from 0.5 at the neutrophil identity sites against the cell's own reference, tared on same-run references (DEV-DIRECTION-02)
`--dev-trace`	`dev_stages.trace_cell`	`dev_stages.py`	3b: another blood group in an isolated-neutrophil specimen, called above 3 x this array's own standard error (DEV-TOOLKIT-ADDED-02)
`--dev-foreign`	`dev_stages.foreign_cell`	`dev_stages.py`	3c: a non-blood template in whole blood by NNLS with the blood groups, called above 3 x this array's own standard error (DEV-TOOLKIT-ADDED-02)
`--dev-brightness`	`dev_stages.brightness`	`dev_stages.py`	11b: 95 % interval on Met-A from the v3 per-site uncertainty (this array's own fixed-site noise), by Monte Carlo (DEV-TOOLKIT-ADDED-02)
`--dev-nilc`	`dev_stages.nilc_e`	`dev_stages.py`	Stage 4 NILC-e composition on the atlas_e templates (DEV-NILC-01 method), 8-group sums and the noise per fraction
`--dev-atlas-e`	`dev_stages.atlas_e`	`dev_stages.py`	Stage 3 atlas_e composition (12 array-measured circulating atlas v2 cells, DEV-ATLAS-EPIC-02 method), 8-group sums
`--dev-percell-b`	`dev_stages.percell_b`	`dev_stages.py`	Stage 5 B-cell Met-A against the composition-matched expectation on the development B-cell floor (DEV-PERCELL-01 method)
`--dev-sky`	`dev_stages.sky`	`dev_stages.py`	Residual sky with the within-chromosome block-shuffle null and the look-elsewhere statistic by simulation (DEV-SKY-02)
`--dev-epic-v2`	`dev_stages.epicv2_calibrate`	`dev_stages.py`	EPIC v2 IDAT pair -> beta Series on EPIC v1 probe names (replicate v2 probes of one CpG averaged). SeSAMe openSesame(prep), pOOBAH-masked

The toolkit (chain/TOOLKIT.md) - built; each stage wired only after its check passed

stage (SOP 2b)	name	module	status
3	Atlas deconvolution	`chain/deconv_v2.py`	toolkit: not yet wired into chain v3
3b	Trace-cell detection	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
3c	Foreign-cell detection	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
4	NILC component separation	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
4	NILC helper: lineage splitter	`chain/Lineage_Splitter/lineage_splitter.py`	toolkit: not yet wired into chain v3
10	Directional decomposition	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
11	Sky map	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
11b	Surface brightness	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
12	Sky statistics	`chain/sky_statistics.py`	toolkit: not yet wired into chain v3
12b	Difference map	`chain/serial_mode.py`	wired into chain v3 (optional input; see chain/TOOLKIT.md)

stage (SOP 2b)	name	module	status
5	Met-A per cell (B cells)	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
8	Self-tare on type II fixed sites	`chain/dev_stages.py`	wired into chain v3: always runs (Stage T step 1); see chain/TOOLKIT.md
1	EPIC v2 calibration	`chain/dev_stages.py`	toolkit: not yet wired into chain v3

Sources

sop/MethylPhys_CPG_SOP_v4.md sha256 241b3c120004

manual/OM_v4_chain.md sha256 25f6561cb3ec

doors/CHAIN_SEQUENCE.md sha256 f805c1c60e8f